

13.1 Universal Gravitation

Question Paper

Course	CIEA Level Physics
Section	13. Gravitational Fields
Topic	13.1 Universal Gravitation
Difficulty	Medium

Time allowed: 40

Score: /32

Percentage: /100

Question 1a

Explain how the force(s) on a satellite can result in the satellite being in a circular orbit around a planet.

[2 marks]

Question 1b

The Earth and the Moon may be considered to be uniform spheres that are isolated in space. The Earth has mass M , radius R and mean density ρ .

The Moon, mass m , is in a circular orbit about the Earth with radius nR , as illustrated in Fig. 1.1.

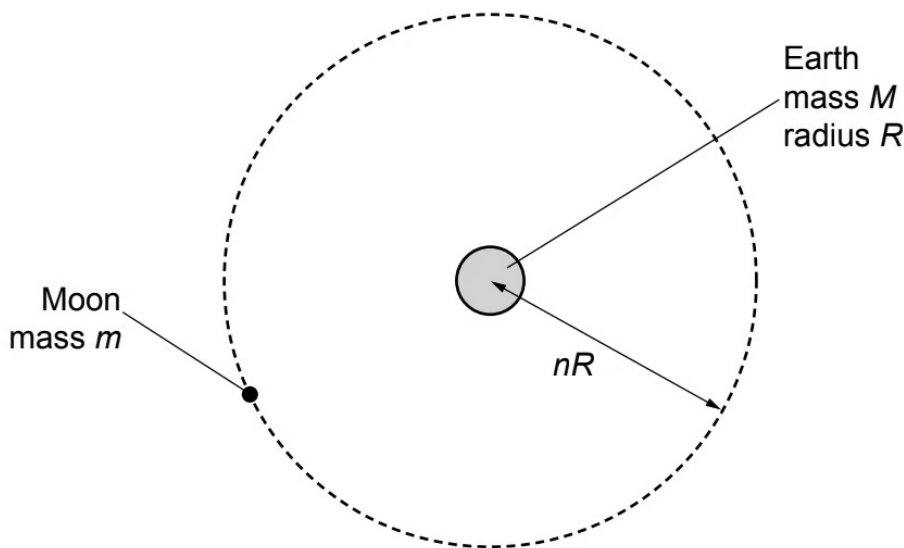


Fig. 1.1

The Moon makes one complete orbit of the Earth in time T .

Show that the mean density ρ of the Earth is given by the expression

$$\rho = \frac{3\pi n^3}{GT^2}$$

where G is the gravitational constant.

[4 marks]

Question 1c

The radius R of the Earth is 6.38×10^6 m and the distance between the centre of the Earth and the centre of the Moon is 3.84×10^8 m.

The period T of the orbit of the Moon about the Earth is 27.3 days.

Use the expression in **(b)** to calculate ρ .

$$\rho = \dots\dots\dots \text{kg m}^{-3}$$

[3 marks]**Question 2a**

Mars may be considered to be a uniform sphere of radius 3400 km with its mass M concentrated at its centre. A stone of mass 3.67 kg rests on top of Olympus Mons, the largest volcano on Mars, which is 25 km high.

The gravitational force on the stone is 13.3 N.

Mars spins on its axis with a period of 1 day and 37 mins.

Calculate M .

[2 marks]

Question 2b

Determine the force required to maintain the stone in its circular path.

[3 marks]

Question 2c

Derive the equation

$$g = \frac{GM}{r^2}$$

where G is Newton's gravitational constant, M is the mass and r is the distance from the mass for the gravitational field strength due to a point mass.

[2 marks]

Question 2d

Olympus Mons is an active volcano.

Calculate the acceleration due to gravity of the stone if it was thrown upwards to twice the height of Olympus Mons.

$a = \dots\dots\dots \text{m s}^{-2}$

[2 marks]

Question 3a

Explain

(i)

What is meant by a *geostationary orbit*.

[3]

(ii)

Why it is preferable to have a satellite in this orbit for TV and telephone signals

[2]

[5 marks]**Question 3b**A satellite of mass m is in orbit around a planet.The mass M of the planet may be considered to be concentrated at its centre.Show that the time period T of the orbit of the satellite is given by the expression

$$T^2 = \left(\frac{4\pi^2 R^3}{GM} \right)$$

where R is the radius of the orbit of the satellite and G is the gravitational constant.

Explain your working.

[4 marks]

Question 3c

The radius of a geostationary orbit around Earth is 4.2×10^7 m.

Calculate the mass of the Earth.

[2 marks]

Question 3d

The radius of the Earth is 6400 km.

Determine the ratio

$$\frac{\text{gravitational field strength of satellite on the Earth's surface}}{\text{gravitational field strength of the satellite at the geostationary orbit}}$$

[3 marks]



Model Answers: Medium

1a

(a) A satellite can be in a circular orbit around a planet due to:

- The gravitational force (of attraction between the satellite and the planet); [1 mark]
- Which causes a centripetal acceleration (of the satellite around the planet); [1 mark]

[Total: 2 marks]

1b

(b) Show that the mean density ρ of the Earth is given by the expression $\rho = \frac{3\pi n^3}{GT^2}$:

List the known quantities:

- Mass of moon = m
- Radius of moon's orbit = nR
- Mass of Earth = M
- Radius of Earth = R

Determine an expression for M in terms of ρ and R :

- $\rho = \frac{M}{V}$
- $V = \frac{4}{3}\pi R^3$
- $M = \frac{4}{3}\pi R^3 \times \rho$ [1 mark]

Consider the centripetal and gravitational forces acting on the moon from the Earth:

- $F_{centripetal} = mnR\omega^2$
- $F_{gravitational} = \frac{GMm}{(nR)^2}$
- $\cancel{m}nR\omega^2 = \frac{GM\cancel{m}}{(nR)^2}$
- $nR\omega^2 = \frac{GM}{(nR)^2}$ [1 mark]

Substitute the equation for angular velocity ω into the equation and rearrange for M :

- $\omega = \frac{2\pi}{T}$ [1 mark]
- $nR\left(\frac{2\pi}{T}\right)^2 = \frac{GM}{(nR)^2}$
- $M = \frac{4\pi^2 n^3 R^3}{GT^2}$

Substitute in the expression for M and rearrange for ρ :

- $\frac{4}{3}\pi R^3 \times \rho = \frac{4\pi^2 n^3 R^3}{GT^2}$
- $\rho = \frac{\cancel{4}\pi\cancel{n}^3\cancel{R}^3}{GT^2} \times \frac{3}{\cancel{4}\pi\cancel{R}^3}$
- $\rho = \frac{3\pi n^3}{GT^2}$ [1 mark]

[Total: 4 marks]

- Start with the main equation for **density**
- Then, figure out what you know about M and V
- The question tells us that $F_{centripetal} = F_{gravitational}$ so we know we can **combine**

these equations

- Circular motion states that

$$\text{Angular velocity } \omega = \frac{\text{distance}}{\text{time}} = \frac{\text{circumference of orbit}}{\text{time period}}$$

1c

(c) Use the expression in **(b)** to calculate ρ :

List the known quantities:

- Radius of Earth, $R = 6.38 \times 10^6 \text{ m}$
- Radius of Moon's orbit, $nR = 3.84 \times 10^8 \text{ m}$
- Period of Moon's orbit around the Earth, $T = 27.3 \text{ days} = (27.3 \times 24 \times 3600) \text{ s}$
- Gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

Calculate the value of n in the radius of the Moon's orbit:

$$\bullet \quad n = \frac{(nR)}{R} = \frac{(3.84 \times 10^8)}{6.38 \times 10^6} = 60.19 \text{ [1 mark]}$$

Substitute the known quantities into the expression for density, ρ :

$$\bullet \quad \rho = \frac{3\pi n^3}{GT^2}$$

$$\bullet \quad \rho = \frac{3\pi \times (60.19)^3}{(6.67 \times 10^{-11})(27.3 \times 24 \times 3600)^2} \text{ [1 mark]}$$

$$\bullet \quad \rho = 5538 \text{ kg m}^{-3} = 5.54 \times 10^3 \text{ kg m}^{-3} \text{ [1 mark]}$$

[Total: 3 marks]

- Don't forget to convert time to **seconds**
- You do **know** n , as you have the information for nR and R

2a

(a) Calculate M

List the known quantities:

- Mass of stone, $m = 3.67 \text{ kg}$
- Radius of Mars, $R = 3400 \text{ km} = 3400 \times 10^3 \text{ m}$
- Height of Olympus Mons, $h = 25 \text{ km} = 25 \times 10^3 \text{ m}$
- Gravitational force, $F = 13.3 \text{ N}$

Use Newton's law of gravitation to determine an equation for M :

$$\bullet \quad F = \frac{GM_1M_2}{r^2} = \frac{GMm}{(R + h)^2}$$

$$\bullet \quad M = \frac{F(R + h)^2}{Gm}$$

Substitute values into the mass equation:

$$\bullet \quad M = \frac{13.3 \times ((3400 + 25) \times 10^3)^2}{(6.67 \times 10^{-11}) \times 3.67} \quad [1 \text{ mark}]$$

$$\bullet \quad M = 6.374 \times 10^{23} = 6.4 \times 10^{23} \text{ kg} \quad [1 \text{ mark}]$$

[Total: 2 marks]

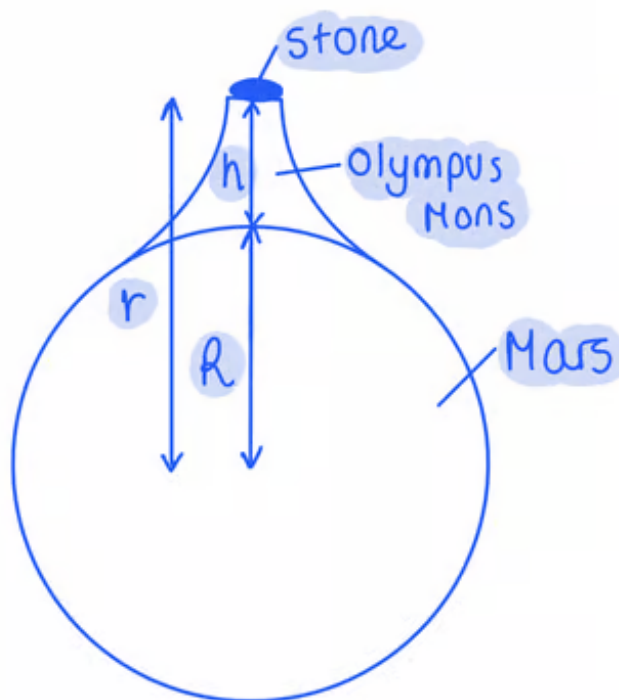
Remember that r is the separation between the **centre** of both the masses. In this case, since the stone is on top of Olympus Mons, we need to take the height of the volcano as well.



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[Total: 2 marks]

Remember that r is the separation between the **centre** of both the masses. In this case, since the stone is on top of Olympus Mons, we need to take the height of the volcano as well.



2b

(b) The force required to maintain the stone in its circular path is:

List the known quantities:

- Mass of stone, $m = 3.67 \text{ kg}$
- Radius of Mars, $R = 3400 \text{ km} = 3400 \times 10^3 \text{ m}$
- Height of Olympus Mons, $h = 25 \text{ km} = 25 \times 10^3 \text{ m}$
- Time period of the stone, $T = 1 \text{ day and } 37 \text{ mins}$

Calculate the time period T in seconds:

- 1 day and 37 mins = $(24 \times 60 \times 60) + (37 \times 60) = 88620 \text{ s}$ [1 mark]

Determine the equation for the centripetal force equation, F :

- $F = \frac{mv^2}{r}$ AND $v = \omega r = \frac{2\pi r}{T}$ [1 mark]

- $F = \frac{m\left(\frac{2\pi r}{T}\right)^2}{r} = \frac{m \times 4\pi^2 r}{T^2}$

Substitute in the values for F :

- $F = \frac{3.67 \times 4\pi^2 \times ((3400 \times 10^3) + (25 \times 10^3))}{(88620)^2} = 6.3 \times 10^{-2} \text{ N}$ [1 mark]

[Total: 3 marks]

2c

(c) Derive the equation $g = \frac{GM}{r^2}$

Determine the relevant equations:

- Newton's law of gravitational states that $F = \frac{GM_1 M_2}{r^2} = \frac{GMm}{r^2}$

AND

- The gravitational field strength is defined as $g = \frac{F}{m}$ [1 mark]

Combine the equations:

- $g = \frac{\frac{GMm}{r^2}}{m} = \frac{GMm}{r^2 m} = \frac{GM}{r^2}$ [1 mark]

[Total: 2 marks]

The difference between M and m is very important! M is the mass of the object creating the gravitational field whilst m is the mass of the object you are calculating the field strength of at that point. Therefore, $g = \frac{F}{M}$ would be incorrect.

2d

(d) Calculate the acceleration due to gravity of the stone if it was thrown upwards to twice the height of Olympus Mons

List the known quantities:

- Mass of stone, $m = 3.67 \text{ kg}$
- Radius of Mars, $R = 3400 \text{ km} = 3400 \times 10^3 \text{ m}$
- Height of Olympus Mons, $h = 25 \text{ km} = 25 \times 10^3 \text{ m}$
- Mass of Mars, $M = 6.374 \times 10^{23} \text{ kg}$ (from part a)

Calculate the gravitational field strength (acceleration due to gravity), g :

- $g = \frac{GM}{r^2} = \frac{(6.67 \times 10^{-11}) \times (6.74 \times 10^{23})}{((3400 \times 10^3) + (25 \times 10^3) + (50 \times 10^3))^2}$ [1 mark]
- $g = a = 3.52 \text{ N kg}^{-1} = 3.5 \text{ m s}^{-2}$ [1 mark]

[Total: 2 marks]

Acceleration due to gravity and the gravitational field strength is the **same thing**. This means that g can either be written in units of N kg^{-1} or m s^{-2} .

3a

(a)

(i) A geostationary orbit is one that is:

- Above the equator; [1 mark]
- Moves from east to west / in the same direction as the Earth's spin; [1 mark]
- Has a period of 24 hours / has the same time period to spin as the Earth; [1 mark]

(ii) It is preferable to have a satellite in this orbit for TV and telephone signals because:

- The satellite would remain in a fixed position above the Earth's surface; [1 mark]
- (Therefore), you wouldn't need to alter the angle / move the receiver; [1 mark]

[Total: 5 marks]

3b

(b) The time period T of the orbit of the satellite is given by the expression

$$T^2 = \left(\frac{4\pi^2 R^3}{GM} \right) \text{ because:}$$

Equate the centripetal and gravitational force:

- The gravitational force provides the centripetal force on the satellite; [1 mark]
- $F = \frac{GMm}{R^2} = mR\omega^2$ [1 mark]
- $\omega = \frac{2\pi}{T}$
- $\frac{GM}{R^2} = R\omega^2 = R\left(\frac{2\pi}{T}\right)^2$ [1 mark]

Rearrange to make T^2 the subject:

- $\frac{GM}{R^2} = R\left(\frac{2\pi}{T}\right)^2 = R\left(\frac{4\pi^2}{T^2}\right)$
- $GM = R^3\left(\frac{4\pi^2}{T^2}\right)$
- $T^2 = \frac{R^3 4\pi^2}{GM}$ [1 mark]

[Total: 4 marks]

Since this question specifically asks you to 'explain your working' - don't just jump

straight into equations. State **why** you are equating centripetal force and gravitational force for a mark.



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- $GM = R^3\left(\frac{4\pi^2}{T^2}\right)$
- $T^2 = \frac{R^3 4\pi^2}{GM}$ [1 mark]

[Total: 4 marks]

Since this question specifically asks you to 'explain your working' - don't just jump straight into equations. State **why** you are equating centripetal force and gravitational force for a mark.

Make sure you lay out any algebra clearly for the examiner. You want to make it easy for them to follow through with your working. This will also help you determine where something may have gone wrong in your final answer if you do not get the required equation.

3c

(c) The mass of the Earth is:

List the known quantities:

- Time period of a geostationary orbit, $T = 24 \text{ hours} = 24 \times 60 \times 60 = 86\,400 \text{ s}$
- Radius of a geostationary orbit around Earth, $R = 4.2 \times 10^7 \text{ m}$

Rearrange the equation from part (c) for the mass M :

- $M = \frac{R^3 4\pi^2}{GT^2}$

Substitute in the values:

- $M = \frac{(4.2 \times 10^7)^3 \times 4\pi^2}{(6.67 \times 10^{-11}) \times (86400)^2}$ [1 mark]
- $M = 5.874 \times 10^{24} = 5.9 \times 10^{24} \text{ kg}$ [1 mark]

[Total: 2 marks]

3d

(d) The ratio $\frac{\text{gravitational field strength of satellite on the Earth's surface}}{\text{gravitational field strength of the satellite at the geostationary orbit}}$ is:

List the known quantities:

- Radius of the Earth, $R = 6400 \text{ km} = 6400 \times 10^3 \text{ m}$
- Radius of the geostationary orbit, $h = 4.2 \times 10^7 \text{ m}$

Determine an equation with the ratio of the orbits:

- $g_1 = \frac{GM}{R^2}$ and $g_2 = \frac{GM}{h^2}$
- $\frac{g_1}{g_2} = \frac{\frac{GM}{R^2}}{\frac{GM}{h^2}} = \frac{h^2}{R^2}$ [1 mark]

Substitute the values into the ratio equation:

- $\frac{g_1}{g_2} = \frac{(4.2 \times 10^7)^2}{(6400 \times 10^3)^2}$ [1 mark]
- $\frac{g_1}{g_2} = 43$ [1 mark]

[Total: 3 marks]

You could of course do this question by calculating g at both positions of the satellite using $g = \frac{GM}{r^2}$, and you would still receive full marks. However, there's a lot more room for calculation errors with large powers that could go wrong. The method in this mark scheme recognises that G and M are going to cancel anyways, but you must be confident with dividing fractions to achieve this.

